Mar & Kuku Production Studio — MVP Guide & Corrected Roadmap (v003)

Updated: 2026-10-22 • Workspace: `/Users/yonassayfu/Projects/mar-kuku-production-studio` • Verified: M0 CH01 + Live 3-Agent Browser Consultation Workflow (ChatGPT + Gemini + Manus)

1. Executive Summary: Completed vs. Partial vs. Remaining Work

This updated guide corrects all earlier readiness claims identified in `codex-status-v002.md` and documents the live, resumable **Idea → Three-Agent Browser Consultation → Saved Script** workflow executed through Studio's own chat.

Category	Component / Workflow	Status	Verified Evidence
Completed	**M0 CH01 Technical Proof & Native Palmier Assembly**	VERIFIED	`E02_CH01_Technical_Proof_v001.palmier` (5 tracks, 16 clips, 10 markers verified on disk & via reopen) and `E02_CH01_Technical_Proof_1080p30_v001.mp4` (907 frames, video=`30.233333`s, audio=`30.200000`s vs `30.220000`s source WAV, `diff=0.020`s). All 279/279 production files in `DoroNesh_2D_Animation` remain byte-identical.
Completed	**Real 3-Agent Browser Consultation via Studio Chat (ChatGPT + Gemini + Manus)**	VERIFIED	One natural-language chat instruction drives managed Chrome (port 9222, persistent profile `scratch/ch01-proof/browser-profile`) sequentially across: (1) **ChatGPT** (`https://chatgpt.com/uc/6ada036c-7d6c-83ea-93d8-22f8a413d90c`, Master Producer), (2) **Google Gemini** (`https://gemini.google.com/app`, Reviewer), (3) **Manus** (`https://manus.im/app/xVApR3Hc7uyqZAMjvXx48B`, authenticated Specialist task), and (4) **ChatGPT Synthesis** (`https://chatgpt.com/uc/6ada03a5-9488-83ea-8e73-827efbd63964`), saving `proposed_script_v001.md`, `proposed_script_v002.md`, `proposed_script.md`, and `output_manifest.json`.
Completed	**Stop / Resume Idempotency & Follow-Up Correction**	VERIFIED	Shared `threading.Event` cancellation across Browser, Provider, Mac, and Workflow controllers. Stopping after Step 1 and typing `Resume the workflow` skips Step 1 (`submission_count == 1`) and finishes Steps 2–4. Follow-up chat correction saves `proposed_script_v002.md` without re-submitting Steps 1–4.
Completed	**Keychain Secret CLI (`mk-studio set-secret`) & OpenAI REST Client**	VERIFIED	`.venv/bin/mk-studio set-secret openai_api --value sk-...` stores secrets in macOS Keychain (`/usr/bin/security`) and enables real HTTPS calls to `https://api.openai.com/v1/responses`.
Partial	**Mac Desktop & Application Control (`MacController`)**	PARTIAL	Permissions (`AXIsProcessTrusted=true`, `system_events=true`, `screen_capture=true`), window inventory, screen capture, app focus, and basic AppleScript keystroke/click work. Full coordinate mouse control, menu bar traversal, arbitrary window targeting, and rich UI postcondition verification remain to be built.
Partial	**Additional AI Providers**	PARTIAL	`chatgpt_web`, `gemini_web`, `manus_web`, and `codex_cli` work live. `openai_api` awaits owner API key via `set-`

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			`secret`; `gemini_cli` is blocked on local Google account tier; DeepSeek web adapter is not yet built.
Remaining	**Asset Generation → Full 7-Chapter Edit → YouTube Studio Upload**	REMAINING	Converting new scripts into generated voice/image assets, extending Palmier scene templates across CH02–CH07, and automating YouTube Studio upload/metadata remain future milestones.

2. Corrections to Previous Readiness Claims

- **Local Sandbox Fixture Explicitly Labeled:** `/browser-sandbox` and the `Sandbox Fixture` button are explicitly labeled as a deterministic local DOM & file upload/download test fixture (`provider_kind="local_fixture"`), never presented as an external AI provider.
- **Real Keychain CLI & OpenAI REST Client:** Added `mk-studio set-secret <provider_id> [--value ...]` in `src/mk_studio/cli.py` and wired real HTTPS requests in `ProviderRegistry._generate_openai_api`.
- **Unified Stop Across All Subsystems:** `POST /api/stop` and chat `Stop` signal a single shared `threading.Event` checked inside `BrowserController` polling loops, `ProviderRegistry` CLI subprocesses, `MacController` commands, and `StudioWorkflowEngine` step checkpoints.
- **Honest QA Stream Durations & Exit Codes:** `verify_exported_mp4` reports both video duration (`30.233333s`, `907` frames) and audio stream duration (`30.200000s` vs `30.220000s` source WAV, `diff=0.020s`). Nonzero exit codes from `verify_ch01` or `assemble_ch01` mark the Studio job as `failed`.

3. Step-by-Step Testing Guide (Start From Scratch)

Step 1: Verify Environment & M0 CH01 Technical Proof

```
cd /Users/yonassayfu/Projects/mar-kuku-production-studio
.venv/bin/mk-studio doctor
.venv/bin/mk-studio verify-ch01
.venv/bin/pytest -q
```

Step 2: Launch the Mar & Kuku Studio Web App

```
.venv/bin/mk-studio serve --host 127.0.0.1 --port 8765
# Open http://127.0.0.1:8765 in your browser
```

Step 3: Run the Real 3-Agent Consultation Workflow Through Studio Chat

In the Studio Chat composer at `http://127.0.0.1:8765` (or by clicking **Run 3-Agent Script Workflow**), send this single instruction:

```
Research the next Mar and Kuku episode, consult Gemini and Manus, and save the proposed script.
```

Watch the managed Chrome window and the left-hand **3-Agent Script Workflow** card as Studio executes:

- **Step 1 — ChatGPT Master Plan (`https://chatgpt.com/`):** Generates episode concept, Reviewer prompt for Gemini, and Specialist TTS timing task for Manus. Saves `01_chatgpt_master_plan.json` & `.md`.
- **Step 2 — Google Gemini Reviewer (`https://gemini.google.com/app`):** Reviews counting clarity, hard pose swaps, and 4-second hold. Saves `02_gemini_review.json` & `.md`.
- **Step 3 — Manus Specialist (`https://manus.im/app`):** Submits the TTS pacing & frame-boundary task to your logged-in Manus session, waits for `Task completed`, and extracts the 110–120 WPM & 120-frame hold specification into `03_manus_specialist.json` & `.md`.

- **Step 4 — ChatGPT Master Synthesis (`https://chatgpt.com/`):** Synthesizes all three agents' outputs into `proposed_script_v001.md`, `proposed_script.md`, and `output_manifest.json` inside `scratch/ch01-proof/episodes/EP03_COUNTING_FLOWERS/`.

Step 4: Test Follow-Up Correction, Stop/Resume, and Owner Takeover

```
# 1. Apply a natural-language follow-up correction in Studio Chat:
Follow-up correction: Revise the proposed script so Kuku giggles 'One! Two! Three purple petals!' right
after the 4-second response hold at frame 621.

# 2. Stop an in-flight workflow and resume without repeating completed submissions:
Stop
Resume the workflow

# 3. Optional: Store an OpenAI API key in macOS Keychain for direct REST calls:
.venv/bin/mk-studio set-secret openai_api --value sk-...
```

4. Verified Episode Folder Artifacts (`EP03_COUNTING_FLOWERS`)

```
scratch/ch01-proof/episodes/EP03_COUNTING_FLOWERS/
  ├── output_manifest.json             # Checkpoint state, live URLs, SHA-256s, submission_count=1 per
step
  ├── 01_chatgpt_master_plan.json/.md  # ChatGPT Master Producer concept & specialist prompts
  ├── 02_gemini_review.json/.md        # Google Gemini Reviewer critique
  ├── 03_manus_specialist.json/.md     # Authenticated Manus specialist TTS & 120-frame hold response
  ├── 04_chatgpt_synthesis.json        # ChatGPT synthesized script v001 payload
  ├── 05_followup_correction_v002.json # Follow-up correction payload (ChatGPT + Gemini confirmation)
  ├── proposed_script_v001.md          # Synthesized script v001
  ├── proposed_script_v002.md          # Revised script v002 incorporating owner correction
  └── proposed_script.md               # Canonical latest script pointer (v002)
```